INCH-POUND

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DETAIL SPECIFICATION

INITIATOR, PROPELLANT ACTUATED, JAU-27/A (MF72)

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the JAU-27/A (MF72) propellant actuated initiator, referred to herein as "initiator".

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-757 - Lead Styphnate, Normal

MIL-P-20444 - Primer, Percussion, M42 Parts for and Loading, Assembling and Packing

MIL-D-21625 - Design and Evaluation of Cartridges for Cartridge Actuated Devices

MIL-DTL-46225 - Lead Azide RD-1333

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-464 - Electromagnetic Environmental Effects Requirements for Systems

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SURFACE WARFARE CENTER INDIAN HEAD (NSWC IHD) (CAGE CODE 76BE6) SPECIFICATIONS

WS 5003 - HNS Explosive

NAVAL AIR SYSTEMS COMMAND (NAVAIR) (CAGE CODE 30003) DRAWINGS

841AS450-3 - Test Tip, Shielded Mild Detonating Cord

850AS131 - Initiator, Propellant Actuated, JAU-27/A

850AS815 - Vibration Fixture, Initiator, Propellant Actuated, JAU-27/A

850AS818 - Test Fixture, Initiator, Propellant Actuated, JAU-27/A

850AS839 - Witness Plate

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquires, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM E545 - Standard Test Method for Determining Image Quality in Direct Thermal Neutron Radiographic Examination

ASTM E1742/E1742M - Standard Practice for Radiographic Examination

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE AS5604 - Compass; Magnetic, Pilot's Standby

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 (see 6.3).

3.2 Design and construction. The initiator shall be manufactured in accordance with Drawing 850AS131.

3.3 Materials. All materials used in the manufacture of the initiators shall conform to the standard established at technical service release as specified (see 6.2). Any divergence shall require technical evaluation and potential design certification testing.

3.4 Primary components. All explosive materials used in the manufacture of this item are considered primary components. Only one lot of primary components shall be used in each lot of initiators; however, a primary component lot may be used in more than one initiator lot.

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3.4.1 PVU-1/A or M42C1-979 percussion primers. Primers used in the initiator shall be of the type originally approved for service release. The PVU-1/A or the M42C1-979 percussion primer shall be manufactured and tested in accordance with MIL-P-20444. The primers shall be loaded into the initiators within 24 months of new primer acceptance testing or within 12 months of primer recertification testing.

3.4.2 Other explosive materials.

3.4.2.1 Lead styphnate. Lead styphnate shall be manufactured in accordance with MIL-DTL-757.

3.4.2.2 Lead azide. Lead azide shall be manufactured in accordance with MIL-DTL-46225.

3.4.2.3 Hexanitrostilbene (HNS). HNS shall be manufactured in accordance with WS 5003, type II, grade A.

3.5 Initiator production.

3.5.1 Explosive train assembly requirements.

3.5.1.1 Visual. The explosive train assemblies shall not have illegible descriptive markings or serial numbers, dents, deep scratches, or other defects that may prevent proper installation or function when examined in accordance with 4.5.1.1.

3.5.1.2 Leakage. Explosive train assemblies shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second (cm^3/s) of air when tested in accordance with 4.5.1.2.

3.5.1.3 Radiographic requirement. The explosive train shall show proper assembly, presence of propellant and all non-metal parts, properly assembled pyrotechnic and non-metallic materials, and no foreign hydrogenous materials when examined radiographically in accordance with 4.5.1.3.

3.5.1.4 Output gap. Explosive train subassemblies shall detonate the test tip high order when tested as described in 4.5.1.4.

3.5.2 Initiator requirements.

3.5.2.1 Firing pin. The firing pin tip feature shall meet dimensional requirements and be centrally aligned. Protrusion of the firing pin shall be verified during the assembly of each firing mechanism in accordance with 4.5.2.1. The firing pin shall move freely under its own weight through the sleeve in accordance with 4.5.2.2.

3.5.2.2 Firing mechanism. Firing mechanisms from each lot, selected in accordance with 4.5.3, shall meet the requirements listed below when subjected to the tests specified in 4.5.3.1 and 4.5.3.2.

3.5.2.2.1 Non-actuation. Firing mechanisms shall not actuate when a pressure of 400 (+0, -25) pounds per square inch gauge (psig) is applied for 1 minute at the inlet port in accordance with 4.5.3.1.

3.5.2.2.2 Actuation. Firing mechanisms not previously subject to the non-actuation pressure in accordance with 3.5.2.2.1 shall actuate when a pressure of 600 (+25, -0) psig is applied in accordance with 4.5.3.2.

3.5.2.3 Visual. The initiators shall not have inadequate or illegible descriptive markings or serial numbers, dents, deep scratches, or other defects that may prevent proper installation or function when examined in accordance with 4.5.4.

3.5.2.4 Radiographic. Initiators shall show proper assembly, presence of parts, and sealing when examined radiographically in accordance with 4.5.5.

3.5.2.5 Demagnetization. Initiators shall not deflect an approved pilot's standby compass more than 3 degrees in either direction when passed in front of the compass. The initiators shall be tested in accordance with 4.5.6.

3.5.2.6 Electrical bonding. The bonding resistance of the initiator across the electrical bonded joint shall be not greater than 0.0025 ohm when tested in accordance with 4.5.7.

3.5.2.7 Threaded joint torque. The breakaway torque of the initiator shall be capable of withstanding a minimum breakaway torque of 200 inch-pounds without loosening when tested in accordance with 4.5.8.

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3.6 Initiator performance. Initiators shall be capable of meeting the requirements of 3.6.1 through 3.6.4.4.

3.6.1 Vibration. The initiator shall not autoignite, shall be safe to handle, and shall show no evidence of deformation or derangement of components when subjected to the vibration test in accordance with 4.5.9. The initiator shall meet the requirements of 3.5.2.4 and 3.6.4 following vibration testing.

3.6.2 Temperature and humidity (T&H) cycling. The initiator shall not incur damage nor deformation when subjected to T&H cycling in accordance with 4.5.10. The initiators shall meet the requirements of 3.5.2.4 and 3.6.4 after being exposed to T&H cycling.

3.6.3 Shock. The initiators shall not incur damage nor deformation when subjected to the shock test in accordance with 4.5.11. The initiators shall meet the requirements of 3.5.2.4 and 3.6.4 following shock testing.

3.6.4 Ballistic requirements. Initiators shall meet the requirements of 3.6.4.1 through 3.6.4.4 when tested in accordance with 4.5.12.

3.6.4.1 Shear pin actuation pressure. The upper (U) and lower (L) limits for shear pin actuation pressure when fired over the temperature range from -65 to +200 °F shall be U = 600 psig and L = 400 psig when evaluated in accordance with 4.5.12.1.

3.6.4.2 Actuation time. The limit for actuation time shall be not greater than 10 milliseconds when evaluated in accordance with 4.5.12.2.

3.6.4.3 High order detonation. The minimum plate dent for high order detonation of the output shielded mild detonating cord (SMDC) test tip on the aluminum witness plate shall be 0.001 inch when evaluated in accordance with 4.5.12.3.

3.6.4.4 Misfire. There shall be no misfires (see 4.5.12.4).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Production inspection (see 4.3).

c. Lot acceptance inspection (see 4.4).

4.2 First article inspection. Unless otherwise specified (see 6.2), an acceptable first article sample of 54 initiators shall be prepared using the same procedures and methods proposed for the production lot. A first article sample acceptable for functional and environmental testing shall be defined as a sample that has met the applicable requirements of 3.2 and 3.5.2.5 of this specification. The first article inspection shall include the examinations and tests of table I in the order specified. All 54 initiators shall be expended in the tests. Failure of any initiator to pass the inspection of table I shall be cause for rejection of the first article sample.

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TABLE I. First article inspection.

Test Sequence	Requirement	Test	Sample Group			
			I	II	III	IV
Visual	3.5.2.3	4.5.4	6	18	6	24
Radiographic	3.5.2.4	4.5.5	6	18	6	24
Electrical Bonding	3.5.2.6	4.5.7	6	18	6	24
Threaded Joint Torque	3.5.2.7	4.5.8	6	18	6	24
Vibration	3.6.1	4.5.9	6	--	--	--
T&H Cycling	3.6.2	4.5.10	--	12	--	--
Shock	3.6.3	4.5.11	--	--	6	--
Radiographic	3.5.2.4	4.5.5	6	18	6	--
Ballistic	3.6.4	4.5.12				
-65±5 °F			2	6	2	8
70±5 °F			2	6	2	8
200±5 °F			2	6	2	8

4.3 Production inspection. The production inspection of all initiators shall consist of the examinations and tests of table II. Any unit failing to pass the production inspection shall be rejected and removed from the lot of explosive train subassemblies or initiators, where applicable.

TABLE II. Production inspection.

Item	Test	Requirement	Test
Explosive Train	Visual	3.5.1.1	4.5.1.1
Explosive Train	Leakage	3.5.1.2	4.5.1.2
Explosive Train	Radiographic	3.5.1.3	4.5.1.3
Explosive Train	Output Gap	3.5.1.4	4.5.1.4
Initiator	Firing Pin	3.5.2.1	4.5.2
Initiator	Firing Mechanism	3.5.2.2	4.5.3
Initiator	Visual	3.5.2.3	4.5.4
Initiator	Radiographic	3.5.2.4	4.5.5
Initiator	Demagnetization	3.5.2.5	4.5.6

4.4 Lot acceptance inspection. Lot acceptance inspection shall consist of the examinations and tests specified in table III. Failure of any sample initiator to comply with the requirements listed in table III shall be cause for rejection of the lot represented.

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TABLE III. Lot acceptance inspection.

Test	Requirement	Test	Quantity
Visual	3.5.2.3	4.5.4	Test and Retain Sample
Radiographic	3.5.2.4	4.5.5	Test and Retain Sample
Electrical Bonding	3.5.2.6	4.5.7	Test and Retain Sample
Threaded Joint Torque	3.5.2.7	4.5.8	Test and Retain Sample
Ballistic 65±5 °F 70±5 °F 200±5 °F	3.6.4	4.5.12	 ⅓ Test Sample ⅓ Test Sample ⅓ Test Sample

4.4.1 Sample size. A random sample of initiators from each production lot, including the samples retained for investigative purposes, shall be selected in accordance with table IV for lot acceptance sampling. Sample initiators made available for investigative purposes shall not be counted as part of the quantity specified for delivery by the contract or order. Unless otherwise specified (see 6.2), the normal sample plan shall be used. A reduced sample size is recommended after 10 consecutive acceptable lots.

TABLE IV. Initiator lot acceptance sampling.

Lot Size	Normal Sample Size	Reduced Sample Size	Retained Sample Size
2 to 50	9	3	3
51 to 100	12	3	3
101 to 150	21	12	3
151 to 280	33	21	3
281 to 500	51	33	3
501 to 1200	81	51	3

4.5 Examinations and tests.

4.5.1 Explosive train.

4.5.1.1 Visual examination. All explosive train subassemblies shall be visually examined and screened for defects prior to initiator assembly. Explosive train assemblies with inadequate or illegible descriptive markings or serial numbers, dents, deep scratches, or other defects that may prevent proper installation or function shall be rejected and removed from the production lot.

4.5.1.2 Leak test. Each explosive train subassembly shall be leak tested in a dry gas leak tester to determine conformance with 3.5.1.2. Units that exhibit a leak rate in excess of 1×10^{-5} standard cm^3/s air at a pressure differential of 1.0±0.1 atmosphere shall be rejected.

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4.5.1.3 Radiographic examination. Each explosive train assembly shall be examined radiographically. Radiographic examination of explosive train subassemblies shall include X-ray in accordance with ASTM E1742/E1742M and N-ray in accordance with ASTM E545. All explosive train subassemblies shall be identified with serial numbers beginning with 001 prior to examination. The initiators shall be arranged on trays or boards in consecutive numerical order with any missing serial numbers identified on the radiographic plate. The explosive train subassemblies shall be positioned for the most revealing exposure with the long axis perpendicular to the radiographic beam. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4- × 6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, the Department of Defense Identification Code, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests (see 6.2). Any observable imperfections in assembly shall be cause for rejection. Defective assemblies found during radiographic review shall be marked on the radiographic plate and removed from the production lot.

4.5.1.4 Output gap test. Prior to the assembly of the initiators, a random sample of explosive train subassemblies, selected per table V, shall be subjected to output gap testing. Explosive train subassembly explosive output shall be capable of detonating SMDC test tips across an air gap that is four times the maximum production gap of the initiator when fired at 75±15 °F. Failure of any explosive train subassembly to meet the requirement of 3.5.1.4 shall result in rejection of the lot.

TABLE V. Explosive train subassembly sampling.

Lot Size	Sample Size
2 to 50	2
51 to 100	4
101 to 150	6
151 to 280	8
281 to 500	10
501 to 1200	12

4.5.2 Firing pin. The firing pin used in each initiator shall be subjected to the examinations described in 4.5.2.1 and 4.5.2.2 prior to and during assembly of the firing mechanism. Parts not satisfying these requirements shall be discarded.

4.5.2.1 Alignment. Alignment and protrusion of the firing pin striker shall be verified by dimensional inspection and visual examination.

4.5.2.2 Firing pin movement. The firing pin, without the O-ring, shall move freely under its own weight throughout the maximum travel provided for the pin through the sleeve bore in the firing mechanism.

4.5.3 Firing mechanism. From each lot, a number of firing mechanisms shall be selected in accordance with general inspection level I of ASQ/ANSI Z1.4 with the additional requirement that the sample size used for non-actuation and actuation testing shall be not less than eight units. If the sample is not equally divisible, the excess initiator shall be placed in the group to be tested at -65 °F. The firing mechanism shall be mated to an inert explosive train initiator to perform testing.

4.5.3.1 Non-actuation. Half of the samples chosen for this test shall be conditioned for at least 4 hours but not more than 24 hours at -65±5 °F. The remaining half shall be conditioned for at least 4 hours but not more than 24 hours at 200±5 °F. Within 3 minutes after removal from the conditioning chamber, a pressure of 400 (+0, -25) psig shall be gradually applied to the inlet port of each firing mechanism and held for 1 minute. Actuation of any firing mechanism shall result in rejection of the lot.

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4.5.3.2 Actuation. Samples shall be chosen as outlined in paragraph 4.5.3. Half of the samples shall be conditioned to -65±5 °F for at least 4 hours but not more than 24 hours. The other half of the samples shall be conditioned to 200±5 °F for at least 4 hours but not more than 24 hours. Within 3 minutes after removal from the conditioning chamber, a pressure of 600 (+25, -0) psig shall be applied at a rate of 10,000 to 50,000 pounds per square inch per second to the inlet port of each firing mechanism. Failure of any firing mechanism to actuate shall result in rejection of the lot. Proper actuation is defined as the shearing of the shear pin, followed by free travel and bottoming of the firing pin against the head of an inert initiator.

4.5.4 Visual examination. Each assembled initiator shall be visually examined and screened for defects. Assembled initiators with inadequate or illegible descriptive markings or serial numbers, dents, deep scratches, or other defects that may prevent proper installation or function shall be rejected and removed from the production lot.

4.5.5 Radiographic examination. Radiographic examination of initiators shall include X-ray in accordance with ASTM E1742/E1742M. All initiators shall be identified with serial numbers prior to examination. The initiators shall be arranged on trays or boards in consecutive numerical order with any missing serial numbers identified on the radiographic plate. The initiators shall be positioned for the most revealing exposure with the long axis perpendicular to the radiographic beam. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4- × 6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests (see 6.2). Defective initiator assemblies found during radiographic review shall be marked on the radiographic plate and removed from the production lot.

4.5.6 Demagnetization test. Each initiator shall be moved past a pilot's standby compass face, which is in accordance with SAE AS5604, by placing the initiator 6 inches from the compass set in an east-west heading. The initiator and the compass shall be placed at the same height from the floor. The initiators shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis, and moved back past the compass face. This procedure shall be repeated for each 90-degree rotation of the initiator. The initiator shall be acceptable if the requirements of 3.5.2.5 are met.

4.5.7 Electrical bonding test. Sample initiators shall be electrically tested for resistance between the two end ports of the initiator, as shown on figure 1, in accordance with MIL-STD-464. The electrical bonding points shall be free from grease, paint, lacquer, or other high resistance properties. Failure of any initiator to comply with the requirements of 3.5.2.6 shall be cause for rejection of the lot or first article.

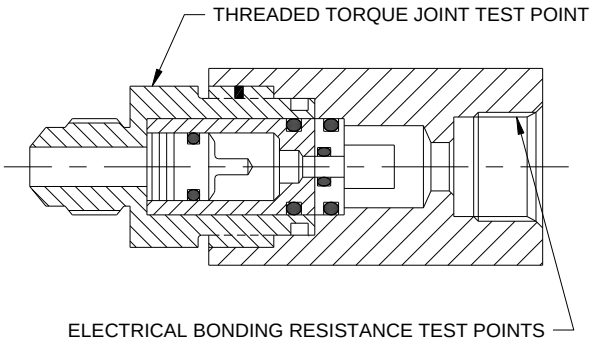


FIGURE 1. Threaded torque joint and electrical bonding resistance test points.

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4.5.8 Threaded joint torque test. Sample initiators shall be tested for the breakaway torque of threaded joints, as shown on figure 1, using a calibrated wrench. Failure of any initiator to comply with the requirements of 3.5.2.7 shall be cause for rejection of the lot or first article.

4.5.9 Vibration tests.

4.5.9.1 Non-gunfire vibration. The test shall be conducted in accordance with the non-gunfire vibration testing of MIL-STD-810, except that for each cycling period, the time shall be divided equally for vibration at -65 °F, 70 °F, and 200 °F. The initiators shall be mounted in the vibration test fixture, which is in accordance with Drawing 850AS815. The non-gunfire vibration test shall be performed on all six initiators. The six initiators shall be mounted on the test fixture at the same time, and one shall be monitored for response during the test.

4.5.9.2 Gunfire vibration. The gunfire vibration test shall be performed on all six initiators on each of the three orthogonal axes as described in 4.5.9.2.1 through 4.5.9.2.5 and shall be conducted at the 70 °F test temperature only. The six initiators shall be mounted on the test fixture at the same time, and one shall be monitored for response during the test.

4.5.9.2.1 Resonance search. A resonance search shall be conducted on each of the six initiators at 70 °F in each of the three orthogonal axes. The frequency sweep shall be made from 50 Hz to 500 Hz and back to 50 Hz in approximately 7.50 minutes to the vibration test levels shown in table VI. The resonance points shall be noted. A resonance point is defined as a frequency that has a transmissibility (output/input) of two or greater. If more than three resonant points are noted from the resonance survey, only the three more severe shall be used for resonance dwell vibration.

TABLE VI. Vibration test levels.

Frequency (Hz)	Acceleration or Amplitude
50 to 54	0.36 inch D.A.
55 to 210	±5 G's
211 to 300	0.00222 inch D.A.
301 to 500	±10 G's
NOTE: G's = gravitational force equivalent; D.A. = double amplitude	

4.5.9.2.2 Resonance dwell. The six initiators shall be vibrated in each of three orthogonal axes at 70 °F for 7.50 minutes at each resonant point (to a maximum of three) obtained during the resonance survey in 4.5.9.2.1. The gunfire vibration test levels are shown in table VI depending upon the number of resonance frequencies determined in the resonance survey per 4.5.9.2. The remainder of the sweep time in accordance with table VII shall be used for vibration cycling per 4.5.9.2.3.

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TABLE VII. Gunfire sinusoidal vibration test schedule.

Fixed Dwell at the Following Center Frequencies (Hz)	Number of Resonances			
	0	1	2	3
	Sweep Time			
	30	22.5	15.0	7.50
	Resonance Dwell Time			
	--	7.5	15.0	22.50
67	7.50	7.50	7.50	7.50
100	7.50	7.50	7.50	7.50
135	7.50	7.50	7.50	7.50
200	15.00	15.00	15.00	15.00
267	7.50	7.50	7.50	7.50
300	7.50	7.50	7.50	7.50
335	7.50	7.50	7.50	7.50
400	15.00	15.00	15.00	15.00
467	7.50	7.50	7.50	7.50
500	7.50	7.50	7.50	7.50
Total Time Each Axis	120.0	120.0	120.0	120.0
NOTE: When an equipment resonance occurs within 5% of a fixed dwell frequency, the fixed dwell period shall be omitted and only the resonance dwell performed. The omitted fixed dwell time shall then be added to the sweep time.				

4.5.9.2.3 Vibration cycling. All six initiators shall be subjected to sinusoidal vibration cycling at 70 °F for a period of time as specified in 4.5.9.2.2 on each of the three orthogonal axes with the frequency varying logarithmically from 50 Hz to 500 Hz and back to 50 Hz in approximately 7.50 minutes. The gunfire vibration test levels shall be as shown in table VI.

4.5.9.2.4 Fixed dwell. After sinusoidal vibration cycling, the six initiators shall be subjected to a series of fixed dwells as shown in table VII. The fixed narrow band dwells shall be performed by sweeping the frequency of ±5 percent of the specified center frequency (e.g., the 100 Hz dwell should be performed from 95 to 105 Hz at the logarithmic sweep rate specified in 4.5.9.2.3).

4.5.9.2.5 Sinusoidal sweep. After completion of the fixed dwell testing, the six initiators shall be subjected to seven sinusoidal vibration sweeps in each of the three orthogonal axes. The frequency shall vary logarithmically from 500 Hz to 2000 Hz and back to 500 Hz. The sweep cycling rate shall be as specified in 4.5.9.2.3 with an applied input of ±10 G's.

4.5.10 T&H cycling test. The T&H cycling test shall be conducted in accordance with MIL-D-21625. Failure of any initiator to comply with the requirements of 3.6.2 shall be cause for rejection of the lot or first article.

4.5.11 Shock test. The shock test shall be conducted in accordance with MIL-D-21625. Failure of any initiator to comply with the requirements of 3.6.3 shall be cause for rejection of the lot or first article.

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4.5.12 Ballistic tests. Sample initiators shall be conditioned for at least 4 hours but not more than 24 hours at the temperatures specified in tables I and III. The initiators shall be removed from the conditioning chamber and test fired within 3 minutes. If any initiator is not fired within 3 minutes after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The initiators shall be fired in the test fixture as shown on figure 2 (see Drawing 850AS818). Instrumentation shall be provided to measure and record the input air pressure (800 [+25, -0] psig), shear pin actuation pressure (in psig), actuation time (in milliseconds), and the high order detonation of the output SMDC test tip on the aluminum witness plate. The initiators shall meet the requirements specified in 3.6.4.

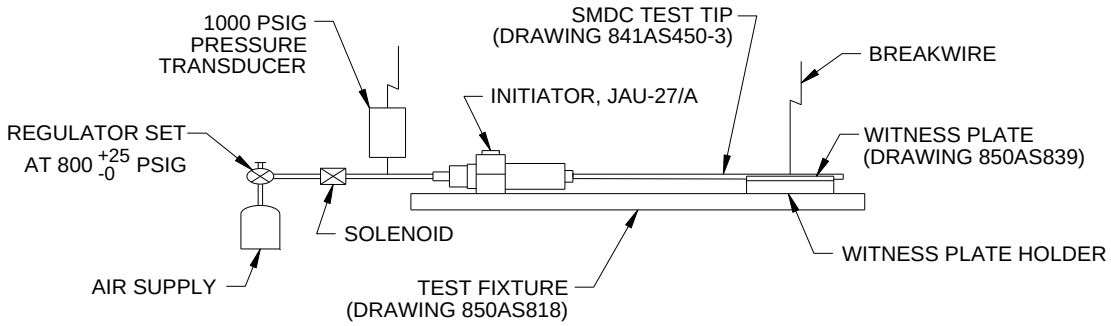


FIGURE 2. Ballistic test setup.

4.5.12.1 Shear pin actuation pressure. The shear pin actuation pressure is the gas pressure required to actuate the gas-operated firing mechanism when a gas pressure of 800 (+25, -0) psig is applied at a rate between 10,000 and 50,000 pounds per square inch per second. The lot shall be acceptable with respect to shear pin actuation pressure if the ballistic sample meets the requirements of 3.6.4.1.

4.5.12.2 Actuation time. The actuation time shall be measured from shear to the detonation of the outlet SMDC test tip. Each initiator shall meet the requirements of 3.6.4.2.

4.5.12.3 High order detonation. High order detonation of the output SMDC test tip shall be demonstrated in a 6061-T6 aluminum witness plate. Each initiator shall meet the requirements of 3.6.4.3.

4.5.12.4 Misfire. An initiator that does not fire under the conditions specified in 4.5.9 shall be considered defective, and the lot shall be considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure. Each initiator shall meet the requirements of 3.6.4.4.

4.5.12.5 Test failure. If test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The propellant actuated initiator converts ballistic gas energy into the explosive energy required to initiate the SMDC used in the FA-18 A/B/C/D/E/F, F-14, TAV-8B, and T-45 aircraft. This propellant actuated initiator is military unique and there is no equivalent commercial application for this design.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether first article inspection is required and, if so, specify the test activity (see 3.1, 4.2, and 6.3).

c. Standard for materials conformance (see 3.3).

d. When alternate materials or methods of manufacture specified on the respective Naval Air Systems Command drawings are used, the bidders' selections must be stated in the proposal (see 3.3).

e. Whether a reduced sample plan is acceptable (see 4.4.1).

f. Directions for shipping radiographs of entire production lot and marriage log (report indicating initiator serial number to initiator serial number), along with the ballistic sample to the activity conducting the lot acceptance tests (see 4.5.1.3 and 4.5.5).

g. Packaging requirements (see 5.1).

h. Whether safety requirements are applicable (see 6.4).

6.3 First article. When first article inspection is required (see 6.2), the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory, and the number of items to be tested (see 4.2). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Any further production prior to notification by the contracting agency of first article sample acceptability is at the contractor's risk.

6.3.1 Bids. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Ammunition and explosive safety. When specified (see 6.2), safety requirements apply to contractors and Government activities.

6.4.1 Contractor's responsibility. The safety requirements of DoD 4145.26-M, DoD Contractor's Safety Manual for Ammunition and Explosives, are applicable when this specification is used for acquisition.

6.4.2 Government activity's responsibility. The safety precaution requirements of NAVSEA OP 5, Ammunition and Explosives Safety Ashore, are applicable when this specification is used as part of a description of work to be accomplished by a Government activity.

6.5 Subject term (key word) listing.

Aircraft, military
Shielded mild detonating cord
SMDC

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2024-016)

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